

GroupTransNet: Group transformer network for RGB-D salient object detection

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ABSTRACT

As an active topic in computer vision, RGB-D salient object detection has witnessed substantial progress. Although the existing methods have achieved appreciable performance, there are still some challenges. The locality of convolutional neural networks requires that the model has a sufficiently deep global receptive field, while the local characteristic represented by transformer with strong globality is always not enough. Besides, the shared information of contextual features tends to be usually overlooked. To address these bottlenecks, we propose a novel group transformer network (GroupTransNet), which is good at learning the long-range dependencies of cross layer features to promote more perfect feature expression between high-level and low-level features. Importantly, we soft group the features of the middle and latter three levels to absorb the semantic information of slightly former level features. Firstly, the input features are adaptively purified by the element-wise operation and sequential attention mechanism. Afterwards, the intermediate features are uniformly fused at different layers, and then processed by several transformers in multiple groups. Finally, the output features are clustered within different classifications and combined with underlying features. Extensive experiments demonstrate the proposed GroupTransNet outperforms the competitors and achieves new state-of-the-art performance.

1. Introduction

The fundamental goal of salient object detection is to automatically identify and segment the most interesting regions or targets of the human eye in a given scene. It has been widely applied in many computer vision tasks, such as image retrieval [1–3], visual tracking [4–7], action recognition [8] and person re-identification [9], and is scalable to serve them as the downstream subtask.

With the popularity of depth sensors in digital cameras and other affordable imaging devices, the acquisition of depth images associated with the corresponding RGB images becomes growing convenient. As a result, the application of RGB-D data is continuously being urged, such as depth map super-resolution [10], depth estimation [11] and many more. Under this tendency, relying solely on a single color image is gradually difficult or unable to meet and attain the demand of advanced detection. In fact, RGB images are capable of offering color clues to gain the texture information of different objects. Meanwhile, depth images facilitate the recording of the distance information of different objects, and provide additional clues to capture the spatial structure and three-dimensional layout. Benefiting from the intrinsic

complementarity of images with different modalities, introducing depth information together with RGB information into salient object detection is theoretically helpful to solve some thorny dilemmas of similar texture, cluttered background or homogeneous objects.

Lots of approaches like early fusion, late fusion and cross level fusion have been proposed recently to focus on the complementary features of RGB modal and depth modal [12–14]. At present, RGB-D salient object detection is concerned by a increasing number of researchers, and corresponding methods are emerging endlessly. These types of methods basically follow a unified two-stage processing step, which first separately extract the RGB and depth features, and then fuse the shallow, deep and cross layer features of the constructed network, respectively. Most of the existing methods on RGB-D saliency detection use convolutional neural networks (CNNs) based on U-Net [15] to progressively encode and decode the features of different layers in deep and shallow layers. Among them, the high-level features from the deep layer are rich in rough semantic information, while the low-level features from the shallow layer are rich in precise detail information. Such a pattern is consistent with the basic mode of encoder-decoder, that

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is, all features are first passed through the encoder and then through the decoder. In this mode, the encoder extracts features from the input image, and the decoder aggregates features to predict the saliency map. However, it is hardly to fully excavate the complex complementary correlations between images with these simple fusion strategies. Most recently, the transformer has caused an uproar in computer vision due to their ability to effectively model the global long-range dependencies of patches in the data expression of images. A natural and reasonable assumption is to leverage transformer to replace or combine CNNs for salient object detection. Although it may be very effective to simply use the former to replace the latter, the integration of the two would be a better option. This is attributed to a consensus that convolutional neural networks are limited in modeling the long-term relationship between features, while transformer fails to capture low-level features. In contrast, they have their own merits reflected in the mining of local features or global features, and the utilization of both local and global features are crucial for detection.

Inspired by TriTransNet [16], it is observed that transformer encoder with shared weight have the potential to more easily catch the common information of multiple features in images. Different from the previous methods, we propose a novel group transformer network (GroupTransNet) to detect salient objects from a pair of cross-modal RGB images and depth images. To the best of our knowledge, the proposed GroupTransNet is the first attempt to express the features of transformer encoder in the data structure of the group. As for our method, the energy weights in the same group pursue the consistency of features, while the energy weights outside different groups pursue the difference of features, which makes the mixed features more recognizable. Instead of treating the features of each level equally, the features of the middle three levels and the last three levels are soft grouped in the proposed method to absorb the semantic information of slightly former level features. Specifically, our GroupTransNet consists of four components, and each of them fulfills its own duties. Firstly, the cross modal features of color modal and depth modal are refined and enhanced by the modal purification module. Then, the scale unification module coordinates the size and interaction of each scale feature. After that, the multiple transformer encoder learns the long-range dependencies of features with high cohesion and low coupling of group. Finally, the ideal features are output in cascade by the cluster integration unit.

In summary, the major contributions of this paper are four-folds:

- The modal purification module is presented, which purifies cross modal features through repeated element-wise refinement and sequential attention enhancement.
- The scale unification module is presented, which makes the size of each scale feature unified and interrelated through the superposition of upsampling and downsampling.
- The multiple transformer encoder is developed, which stacks the grouped transformer bodies with multiple shared energy weights to more specifically express the features, thereby learning the common information of cross modal and each scale features within different groups.
- The cluster integration unit is designed, which realizes the robust and efficient combination of different types of high-level features towards low-level features by exploiting the influence of dislocation deviation of feature levels in each cluster.

The rest of this paper is organized as follows. Section 2 concisely retrospects some relevant works in recent years. Section 3 describes our proposed method and its components in a detailed manner. Extensive experiments are conducted in Section 4. The conclusion is given in Section 5.

2. Related works

In this section, we provide a brief literature review of some recent works related to the topic from two main aspects, including saliency detection and vision transformer.

2.1. Saliency detection

Saliency object detection, also called saliency detection for short, is one of the most important branches in the field of computer vision [17]. So far, researchers have proposed a large number of saliency detection methods [18–25]. These methods mainly design models to aggregate high-level features and low-level features. For instance, Wu et al. [26] utilizes the interaction between the edge module and the detection module to optimize these two tasks at the same time. Su et al. [27] uses boundary to guide encoder and decoder to gradually optimize saliency prediction. Chen et al. [28] integrates low-level appearance features, high-level semantic features and global context features, and generate saliency map in a supervised manner.

In recent years, it has been found that the depth information with three-dimensional layout and spatial structure can assist the color information, and the use of these two kinds of information can improve the detection performance [29]. Therefore, researchers have proposed plenty of RGB-D saliency detection methods [30–38]. According to the fusion form of the two modals, these methods can be divided into early fusion, late fusion and cross level fusion. For instance, Qu et al. [39] presents an interactive mechanism for automatic learning of RGB-D salient object detection. Wang et al. [40] designs two stream convolution neural networks, in which each network extracts features from color or depth pattern and learns the switch map to adaptively fuse the predicted saliency map. Zhang et al. [41] proposes to simulate the uncertainty of human annotation through the network of automatic encoder and generate multiple saliency maps for each input image by sampling in the potential space. Piao et al. [42] realizes the expected control of pixel level depth knowledge transmitted to the color stream by adaptively minimizing the difference between the prediction generated by the depth stream and the color stream. Pang et al. [43] integrates the features of different modals through densely connected structures, and uses their mixed features to generate dynamic filters with different sizes of receptive fields. Ji et al. [44] proposes a new collaborative learning framework to make use of edge, depth and saliency in a more effective way. Fan et al. [45] proposes to implement the filtering of low-quality depth map and cross modal feature learning. Xu et al. [46] develops a weakly supervised RGB-D saliency detection task with only scribble annotations. Cong et al. [47] inserts a refinement middleware structure between the encoder and the decoder for the cross-modality refinement. Zhou et al. [48] applies a multilevel atrous spatial pyramid pooling to hierarchical network modules to obtain multiscale information. Wang et al. [49] integrates a module for filtering low-quality depth maps and conducts the quality enhancement procedure towards these detrimental depth maps.

Although existing deep learning based methods have reported promising results, they fail to harmoniously integrate transformer into CNNs. In other words, CNNs and transformer are not organically merged. Unlike prior works, we thoroughly explore the integration of the two and complement each other under different strategies.

2.2. Vision transformer

Transformer is originally put forward in just a few short years [50]. Once proposed, this technology quickly occupies a dominant position in the field of natural language processing, constantly refreshing records one after another. The advantage of it lies in its ability to handle sequence data well and the strong robustness in modeling global long-range dependencies. Recently, thanks to its success, researchers have extended it into computer vision field and achieved impressive results, thus steadily winning a place.

For instance, Liu et al. [51] presents a hierarchical transformer with shift window scheme. Liu et al. [52] proposes to make the model more effective in transmitting resolution across windows. Chen et al. [53] proposes using transformer to encode the labeled image block in CNNs feature map into an input sequence for extracting global

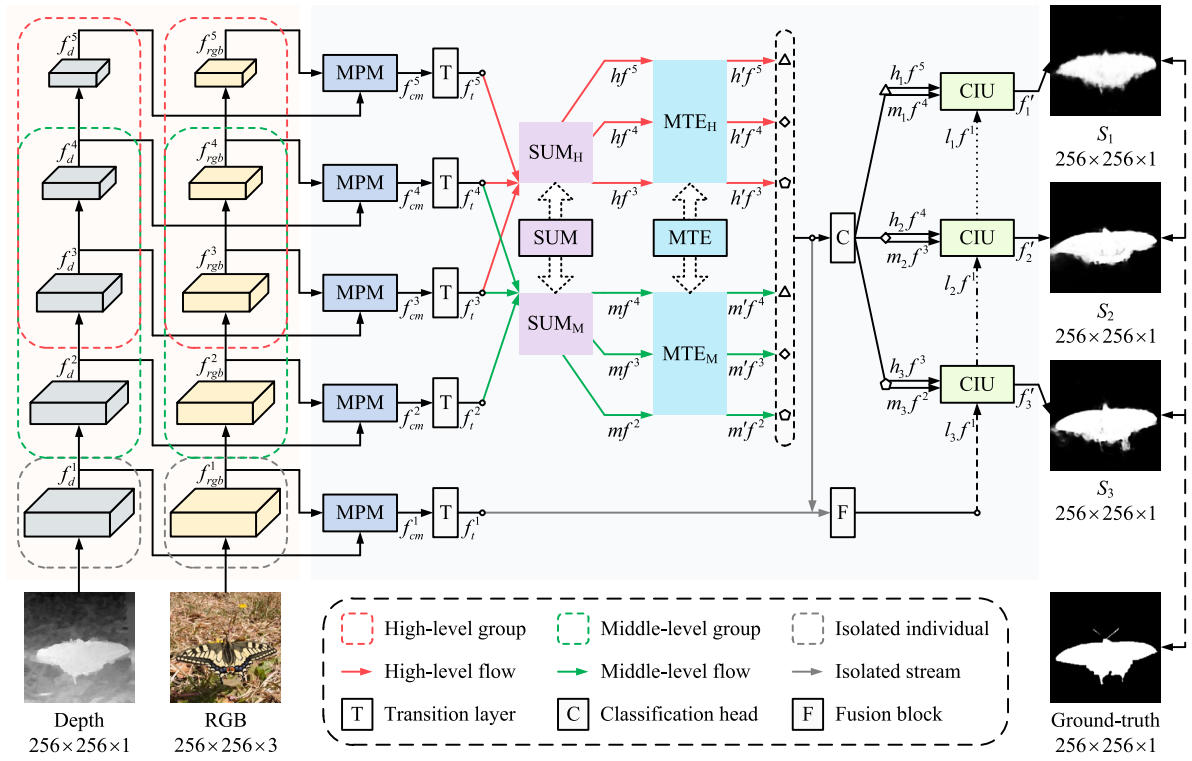


Fig. 1. The overall architecture of our group transformer network (GroupTransNet). There are four core components that are deployed and distributed, namely modal purification module (MPM), scale unification module (SUM), multiple transformer encoder (MTE) and cluster integration unit (CIU).

context and combining the decoder with high-resolution CNNs feature map to achieve accurate positioning. Dosovitskiy et al. [54] interprets the image as a sequence of flat two-dimensional patches, and uses transformer for image classification. Yuan et al. [55] uses a tokens-to-token module to model the local structure, so as to generate multi-scale token features. Liu et al. [56] takes image patches as input and used transformer to propagate global context between image patches. Han et al. [57] regards local patches as visual sentences and further divided them into smaller patches as visual words, in which the attention of each word will be calculated together with other words in a given visual sentence. Li et al. [58] improves the locality mechanism of information exchange in the local area by introducing deep convolution into the feedforward network and adding locality to the visual transformer. Liu et al. [16] proposes the triple transformer embedded module to learn cross layer long-range dependencies to enhance high-level features. Tang et al. [59] proposes to capture significant and common visual patterns from multiple images. Ren et al. [60] proposes a pure transformer based encoder and a hybrid decoder to aggregate the features generated by transformer. Wang et al. [61] introduces a new pixel level focus regularization to guide CNNs feature extraction program, and deeply utilizes the multimodal interaction of color and depth images on coarse and fine scales. Lin et al. [62] alternates attention inner the image patch to capture local information and apply attention between image patches to capture global information. Heidari et al. [63] merges the long-range contextual interactions of the transformer and the local semantic information of CNNs.

Although transformer has been successfully applied in various fields of computer vision, its ability in specific tasks is not stimulated, which is often nested as the independent unit or unrelated combination. Unlike prior works, we take into account group strategy and advocate group transformer for the task of RGB-D saliency object detection.

3. Proposed method

In this section, we explain the proposed group transformer network (GroupTransNet) in detail. Specifically, we first display the network

architecture, then elaborate the proposed components in the network in turn, and finally give the loss function that need to be optimized.

3.1. Architecture overview

As shown in Fig. 1, the overall architecture of the proposed GroupTransNet consists of core components, which are modal purification module (MPM), scale unification module (SUM), multiple transformer encoder (MTE) and cluster integration unit (CIU). From a macro perspective, features at different scales under different modals are empirically grouped throughout the entire pipeline process from start to finish. For these five different scale input features $\{f^i | i = 1, 2, 3, 4, 5\}$ from color modal and depth modal respectively, they are soft divided into two groups $\{f^2, f^3, f^4\}$ and $\{f^3, f^4, f^5\}$, as well as an isolated point $\{f^1\}$. The features learned from these two groups are combined with that from the isolated point in a hierarchical manner to generate output features. First, all input features are purified by MPM to obtain cross modal features. Second, the grouped features are processed by SUM to make the size of each scale feature unified and interrelated. Third, the intermediate features learn the common information in these groups pass through MTE to get more discriminative feature representation. Fourth, the high level features and low level features are integrated to generate output features, which is realized in CIU. Hence, the final saliency map is computed by integrating the obtained feature map. It is worth noting that several plugins are mixed in the whole process, namely the transition layer, classification head and fusion block. Specifically, their responsibilities are to switch the number of channels for features, redirect different types of features into corresponding classes, and cross generate new low-level features from the isolated individual.

3.2. Modal purification module

In RGB-D saliency detection, there are two kinds of materialized expression forms of image information, which are color modal and

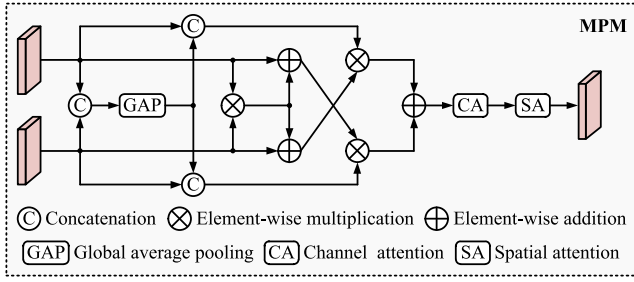


Fig. 2. The schematic diagram of modal purification module (MPM).

depth modal. Specifically, the color modal provides the appearance clues of the image, while the depth modal gives the distance clues of the image. They add useful information for detection in different modes. However, the features of different modals are incompatible to a certain extent, which is caused by the inherent differences of modals. If these two modals are simply utilized, such as concatenating the color modal directly with the depth modal, it will bring serious noise to the features. To address one, a modal purification module (MPM) is presented.

The schematic diagram of MPM is shown in Fig. 2. MPM first carries out feature refinement by repeated element-wise operation linkage, and then carries out feature enhancement by sequential attention mechanism support. Among them, the attention mechanism refers to the channel attention mechanism and spatial attention mechanism, which are stated in CBAM [64].

For color feature f_{rgb}^i ($i = 1, 2, 3, 4, 5$) and depth feature f_d^i ($i = 1, 2, 3, 4, 5$), feature refinement and feature enhancement can purify the cross modal features of color modal and depth modal, and obtain the combined feature f_{cm}^i ($i = 1, 2, 3, 4, 5$). To be specific, the refinement process can be defined as:

$$\widetilde{f_{rgb}^i} = Cat(GAP(Cat(f_{rgb}^i, f_d^i)), f_{rgb}^i), \quad (1)$$

$$\widetilde{f_d^i} = Cat(GAP(Cat(f_{rgb}^i, f_d^i)), f_d^i), \quad (2)$$

$$\widetilde{f_{rgb}^i} = ((f_{rgb}^i \otimes \widetilde{f_d^i} \oplus \widetilde{f_d^i}) \otimes \widetilde{f_{rgb}^i}), \quad (3)$$

$$\widetilde{f_d^i} = ((f_{rgb}^i \otimes \widetilde{f_d^i} \oplus \widetilde{f_{rgb}^i}) \otimes \widetilde{f_d^i}), \quad (4)$$

$$\widehat{f^i} = \widetilde{f_{rgb}^i} \oplus \widetilde{f_d^i}, \quad (5)$$

where $Cat(\cdot)$ denotes the concatenation operation, $GAP(\cdot)$ denotes the global average pooling operation, $\otimes$ and $\oplus$ denote the operations of element-wise multiplication and element-wise addition, respectively. The refinement process continues to cross supplement the color features and depth features, refine the depth features with the color features, and then refine the color features with the depth features, so as to iterate repeatedly. In addition, the enhancement process can be defined as:

$$f_{cm}^i = SA(CA(\widehat{f^i})), \quad (6)$$

where $CA(\cdot)$ and $SA(\cdot)$ denote channel attention mechanism and spatial attention mechanism, respectively. The enhancement process feeds into the channel attention mechanism and spatial attention mechanism in turn. It first enhances the features in the channel way, and then enhances the features in the spatial way.

3.3. Scale unification module

In the network, the features of each scale have different resolutions due to several sampling. Apart from that, the semantic information and detail information contained in the features of each scale are very different. On the one hand, features of different sizes cannot be properly handled in a custom enclosure. On the other hand, feature interaction

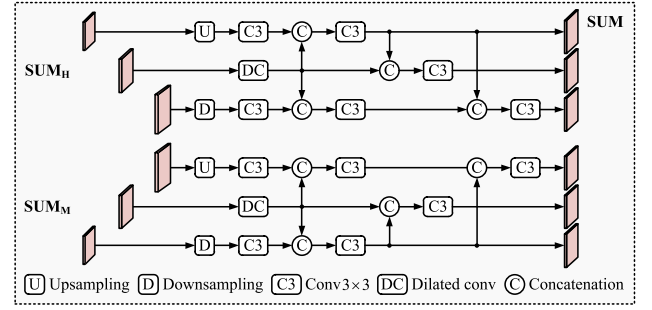


Fig. 3. The schematic diagram of scale unification module (SUM).

at all scales is also very important from beginning to end. In order to solve these problems, a scale unification module (SUM) is presented.

There are actually two versions of SUM, corresponding to the two groups formed by soft partition, SUM_H and SUM_M. Their schematic diagram is shown in Fig. 3. Both of them upsampling or downsampling the higher and lower layers to the scale of the middle layer, respectively, and then a series of symmetrical concatenation operations are carried out.

Before that, the features of all layers f_{cm}^i ($i = 1, 2, 3, 4, 5$) are passed through a transition layer, so that the number of channels from low to high is collectively normalized from 64, 256, 512, 1024 and 2048 to 64, and the transited features f_t^i ($i = 1, 2, 3, 4, 5$) are obtained. The transition layer contains a 3×3 convolution layer and a ReLU activation function.

The features of the higher three layers and the middle three layers are sent to SUM in block shape, in which the features of the higher three layers f_t^i ($i = 3, 4, 5$) are sent to SUM_H and the features of the middle three layers f_t^i ($i = 2, 3, 4$) are sent to SUM_M to obtain the features hf^i ($i = 3, 4, 5$) and mf^i ($i = 2, 3, 4$), respectively. For a raw triplet $\{f_t^{i+1}, f_t^i, f_t^{i-1} | i = 4\}$, SUM_H converts it to a high level triplet $\{hf^{i+1}, hf^i, hf^{i-1} | i = 4\}$. Similarly, SUM_M converts a raw triplet $\{f_t^{i+1}, f_t^i, f_t^{i-1} | i = 3\}$ to a middle level triplet $\{mf^{i+1}, mf^i, mf^{i-1} | i = 3\}$. The two phases of SUM can be defined as:

$$\{hf^{i+1}, hf^i, hf^{i-1}\} = F_{SUM}^H\{f_t^{i+1}, f_t^i, f_t^{i-1}\}, \quad (7)$$

and

$$\{mf^{i+1}, mf^i, mf^{i-1}\} = F_{SUM}^M\{f_t^{i+1}, f_t^i, f_t^{i-1}\}, \quad (8)$$

where F_{SUM}^H and F_{SUM}^M refer to the functions of SUM_H and SUM_M, respectively.

Let both of the three features of the higher three layers and the middle three layers from high to low be expressed as f^h , f^m and f^l , respectively. SUM_H and SUM_M can be defined as:

$$f^h = C_3(Cat(C_3(U p(f^h)), DC(f^m))), \quad (9)$$

$$f^m = C_3(Cat(DC(f^m), f^h)), \quad (10)$$

$$f^l = C_3(Cat(C_3(Cat(C_3(Down(f^l)), DC(f^m))), f^h)), \quad (11)$$

and

$$f^h = C_3(Cat(C_3(Cat(C_3(U p(f^h)), DC(f^m))), f^l)), \quad (12)$$

$$f^m = C_3(Cat(DC(f^m), f^l)), \quad (13)$$

$$f^l = C_3(Cat(C_3(Down(f^l)), DC(f^m))), \quad (14)$$

where $U p(\cdot)$ and $Down(\cdot)$ denote the upsampling and downsampling operations, respectively, $DC(\cdot)$ denotes the dilated convolution operation, $Cat(\cdot)$ denotes the concatenation operation, $C_3(\cdot)$ is a 3×3 convolutional layer.

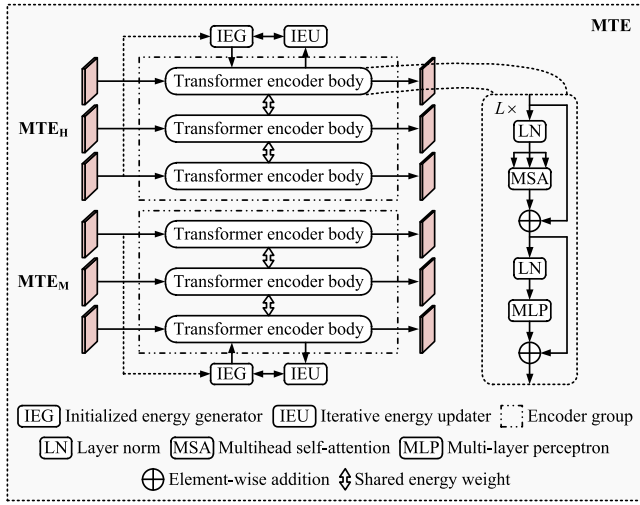


Fig. 4. The schematic diagram of multiple transformer encoder (MTE).

3.4. Multiple transformer encoder

In order to make better use of the advantages of transformer, a multiple transformer encoder (MTE) is developed. It can learn common information from multi-level features. More importantly, it enjoys high cohesion inside different groups and low coupling outside different groups, which makes the learned feature information more discriminative.

The feature $X \in \mathbb{R}^{H \times W}$ is first reshaped into a 2D sequence of flattened patches, namely $X^i \in \mathbb{R}^{P^2}$ ($i = 1, 2, \dots, N$), in which $N = (H \times W)/P^2$ is the number of patches. In this way, the size of each patch is $P \times P$. Then, a trainable linear projection is used to map the patch to the potential D -dimensional embedding space. After that, the positional information of each patch is encoded through learning position embedding and added to patch embedding. The process can be formalized as:

$$z_0 = [X^1 E; X^2 E; \dots; X^N E] + E_{pos}, \quad (15)$$

where $E \in \mathbb{R}^{P^2 \times D}$ denotes the patch embedding operation, $E_{pos} \in \mathbb{R}^{N \times D}$ denotes the position embedding operation.

The standard transformer encoder is composed of L transformer layers assembled alternately. In each layer, multihead self-attention and multi-layer perceptron are two key blocks. The layer norm is embedded before them and the residual connection is embedded after them. The specific process is formalized as:

$$z'_l = MSA(LN(z_{l-1})) + z_{l-1}, \quad l = 1, 2, \dots, L, \quad (16)$$

$$z_l = MLP(LN(z'_l)) + z'_l, \quad l = 1, 2, \dots, L, \quad (17)$$

where $LN(\cdot)$, $MSA(\cdot)$ and $MLP(\cdot)$ denote the layer norm, multihead self-attention and multi-layer perceptron operations, respectively.

In contrast, this multiple transformer encoder attempts to find another way to achieve the synchronous improvement of encoder and decoder by placing different shared weights between two groups of encoders composed of three standard transformer encoders. What is more, the input sizes of the two groups of encoder bodies are different. The input of MTE_H is 16×16 and that of MTE_M is 32×32 . Their outputs are consistent with their respective inputs. It should be noted that 64×64 is not used as input, because the memory to bear is too large, which is generally difficult to achieve.

There are also actually two versions of MTE, corresponding to the two groups formed by soft partition, MTE_H and MTE_M . Their schematic diagram is shown in Fig. 4. They differentiate by grouping, and each encoder group is built by multiple transformer encoder bodies sharing

weight. The transformer encoders used by the two have no other difference except the size of input and output along with the shared energy weight values.

The features $h^i f^i$ ($i = 5, 4, 3$) and $m^i f^i$ ($i = 4, 3, 2$) are passed through two encoders groups with different grid sizes to obtain new features $h' f^i$ ($i = 5, 4, 3$) and $m' f^i$ ($i = 4, 3, 2$). For a high level triplet $\{h^i f^{i+1}, h^i f^i, h^i f^{i-1} | i = 4\}$, MTE_H converts it to an updated triplet $\{h' f^{i+1}, h' f^i, h' f^{i-1} | i = 4\}$. Similarly, MTE_M converts a middle level triplet $\{m^i f^{i+1}, m^i f^i, m^i f^{i-1} | i = 3\}$ to a updated triplet $\{m' f^{i+1}, m' f^i, m' f^{i-1} | i = 3\}$. The two phases of MTE can be defined as:

$$\{h' f^{i+1}, h' f^i, h' f^{i-1}\} = \mathcal{F}_{MTE}^H \{h^i f^{i+1}, h^i f^i, h^i f^{i-1}\}, \quad (18)$$

and

$$\{m' f^{i+1}, m' f^i, m' f^{i-1}\} = \mathcal{F}_{MTE}^M \{m^i f^{i+1}, m^i f^i, m^i f^{i-1}\}, \quad (19)$$

where $\mathcal{F}_{MTE}^H$ and $\mathcal{F}_{MTE}^M$ refer to the functions of MTE_H and MTE_M , respectively.

Let both of the three features of the higher three layers and the middle three layers from high to low be expressed as a triplet $\{f^h, f^m, f^l\}$, respectively. MTE_H and MTE_M can be defined as:

$$\{f^h, f^m, f^l\} = TEB_l(\{f^h, f^m, f^l\}, \eta), \quad (20)$$

and

$$\{f^h, f^m, f^l\} = TEB_s(\{f^h, f^m, f^l\}, \varphi), \quad (21)$$

where $TEB_l(\cdot)$ and $TEB_s(\cdot)$ represent a set of encoders with large and small scales, respectively, η and φ are two different energy weight parameters that act on two different sets of encoders. The energy weight is defined as a special weight that can be learned from the correlation of group features, and it is customized to strive for internal consistency and external difference of the encoder. From the functional perspective, the energy weights here are somewhat similar to the weights in MiniViT [65] and DWNet [66]. Notably, the values of energy weights are determined by the designed energy generator and energy updater. The calculation of energy weight is formulated as:

$$\eta = IEU(IEG(f^h, f^m, f^l), \rho_1), \quad (22)$$

$$\varphi = IEU(IEG(f^h, f^m, f^l), \rho_2), \quad (23)$$

where $IEG(\cdot)$ is the initialized energy generator, $IEU(\cdot)$ is the iterative energy updater, and ρ_i ($i = 1, 2$) are two variable learning factors with values ranging from 0 to 1, which are automatically updated from each iteration. Two energy weights are assigned by the initialized energy generator and changed by the iterative energy updater. The energy generator obtains the fitness of features in the pretrained model and is responsible for randomly initializing a shared parameter at the beginning. The energy updater fine tunes the fitness by leveraging the relevance of features and is responsible for dynamically updating the shared parameter in subsequent iterations. The two that generate energy weights can enable feature sharing and enable adaptive network adjustments.

The two encoders groups share their own weights, and the internal relevance of these features at the high and middle level groups is guaranteed. Meanwhile, the two encoders are irrelevant to each other, which also ensures the external differences of these features at the high and middle level groups.

Subsequently, these features are passed through a classification head together, resulting in a pairwise combination of classification features. In the meantime, they are also dispersed along with the lowest level feature through a fusion block. The fusion block includes a 3×3 convolution layer, a 1×1 convolution block, a batch regularization and a ReLU activation function. It interlaces group features onto the lowest level feature to meet better detail requirements.

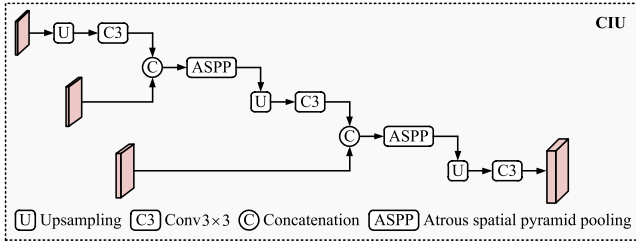


Fig. 5. The schematic diagram of cluster integration unit (CIU).

3.5. Cluster integration module

In order to cascade the features of different layers and even those with recognition in the same layer in an orderly manner, a cluster integration unit (CIU) is designed.

The schematic diagram of CIU is shown in Fig. 5. It is a cascading way to integrate the clustered features in the order from high to low, which is first upsampled by the high-level features and then concatenated with the low-level features.

According to the criteria of interleaving feature differences from two different groups of encoder bodies, the generated features $\{h'f^5, h'f^4, h'f^3\}$ and $\{m'f^4, m'f^3, m'f^2\}$ are converted to features $\{h_1f^5, h_2f^4, h_3f^3\}$ and $\{m_1f^4, m_2f^3, m_3f^2\}$, and then clustered into three classes, namely $C_1 = \{h_1f^5, m_1f^4\}$, $C_2 = \{h_2f^4, m_2f^3\}$ and $C_3 = \{h_3f^3, m_3f^2\}$. In the first class C_1 , h_1f^5 contains the feature information of equal fourth and fifth layers, while m_1f^4 contains the feature information of equal second and fourth layers and more third layers. Therefore, it has rich information from the second layer to the fifth layer, and this property is suitable for biased lossless features. Similarly, in the second class C_2 , h_2f^4 contains the feature information of equal fifth layer and more fourth layer, while m_2f^3 contains the feature information of equal second layer and more third layer. At the same time, in the third class C_3 , h_3f^3 contains the feature information of equal third layer and fifth layer and more fourth layer, m_3f^2 contains equal feature information of the second and third layers. Therefore, they also satisfy the same properties as the first class.

These three classes pay attention to the feature information except the first layer with different emphasis, so it is necessary to combine them with the feature $\{l_1f^1\}$, $\{l_2f^1\}$ and $\{l_3f^1\}$ containing the first layer information, respectively. Additionally, the atrous spatial pyramid pooling (ASPP) [67] is used to better adapt to the aggregation of features at multiple scales. The integration process of all features to be combined can be defined as:

$$f'_i = C_3(U p(\Gamma(Cat(C_3(U p(\Gamma(Cat(C_3(U p(C_{i1})), C_{i2})))), l_i f^1))), (24)$$

where C_{ij} ($i = 1, 2, 3, j = 1, 2$) indicates the j th element in the i th class, $C_3(\cdot)$ is a 3×3 convolutional layer, $\Gamma(\cdot)$ stands for the ASPP operation, $U p(\cdot)$ and $Cat(\cdot)$ denote the upsampling and concatenation operations, respectively.

3.6. Loss function

Assuming that there are three side outputs f'_i ($i = 1, 2, 3$), the predicted saliency map S_i ($i = 1, 2, 3$) from them is produced as:

$$S_i = \text{Sig}(C_1(C_3(U p(f'_i)))), (25)$$

where $U p(\cdot)$ denotes the upsampling operation, $C_3(\cdot)$ and $C_1(\cdot)$ are the 3×3 and 1×1 convolutional layers, respectively, $\text{Sig}(\cdot)$ denotes the Sigmoid function. Note that the Sigmoid function is used to compute the probability for each pixel of being salient or not, which also involves the combination of convolution and ReLU operations.

Similar to TriTransNet [16], the supervised learning of model error also adopts the pixel position aware (PPA) loss [68], denoted by $\mathcal{L}_{ppa}$,

which is the joint loss function by employing the weighted binary cross entropy (wBCE) loss and weighted intersection over union (wIoU) loss. Thus, the total loss function is defined as:

$$\mathcal{L}_{Total} = \sum_{i=1}^3 \mathcal{L}_{ppa}(S_i, G), (26)$$

where G is ground-truth saliency map.

4. Experiments

In this section, we begin by introducing the commonly used datasets and evaluation metrics, as well as training protocols and implementation details. Next, we quantitatively and qualitatively compare the proposed method with the state-of-the-art methods to prove the advantages. Afterwards, we conduct a series of ablation studies to verify the role of each component of our network. In addition, some failure cases is also discussed at the end.

4.1. Experimental setup

4.1.1. Datasets

To measure the performance of the proposed method, we execute validation on six benchmark datasets, including STEREO [69], NJU2K [70], NLPR [71], RGBD135 [72], DUT-RGBD [73] and SIP [45]. These datasets can be publicly downloaded from their official website. STEREO, also called SSB1000, is composed of 1000 stereoscopic image pairs and corresponding ground-truth crawled from the Internet, in which the depth images are estimated from the stereo images. NJU2K consists of 1985 image pairs along with ground-truth collected from the Internet, 3D movies and photographs, in which the depth images are estimated from the stereo images. NLPR contains 1000 image pairs with their ground-truth taken under different illumination conditions, in which the depth images are captured by the Microsoft Kinect. RGBD135, also known as DES, is composed of 135 indoor image pairs and the ground-truth, in which the depth maps are captured by the Microsoft Kinect. DUT-RGBD contains 1200 image pairs with their ground-truth taken in varied real life situations, in which the depth images are captured by the Lytro light field camera. SIP has 929 high-resolution image pairs and corresponding ground-truth of person activities, in which the depth images are captured by the Huawei Meta10.

4.1.2. Evaluation metrics

To measure the performance of the proposed method and other methods, we choose four typical evaluation metrics, including S-measure (S_a) [94], F-measure (F_β) [95], mean absolute error that is abbreviated as MAE ($\mathcal{M}$) [96] and E-measure (E_e) [97]. These metrics are all recognized as highly representative. S-measure calculates the structural similarity between the prediction and ground-truth, which is defined as:

$$S_a = \alpha \times S_o + (1 - \alpha) \times S_r, (27)$$

where S_o and S_r denote the object-aware structural similarity and region-aware structural similarity, respectively, and α is set to 0.5 to assign equal constraints to both. F-measure calculates the harmonic mean of the thresholded precision and recall, which is defined as:

$$F_\beta = \frac{(1 + \beta^2) \times \text{Precision} \times \text{Recall}}{\beta^2 \times \text{Precision} + \text{Recall}}, (28)$$

where β^2 is set to 0.3 to emphasize precision more than recall as suggested in [98], and it need be pointed out that we adopt the average F-measure here. E-measure calculates the local pixel-level error and the global image-level error together, which is defined as:

$$E_e = \frac{1}{W \times H} \sum_{i=1}^W \sum_{j=1}^H \phi(x, y), (29)$$

Table 1

Performance comparison with 27 state-of-the-art methods on six datasets. $\uparrow$ and $\downarrow$ stand for larger and smaller is better, respectively. ‘-’ represents that data is not available. The best results are marked in **bold**.

Methods	STEREO [69]				NJU2K [70]				NLPR [71]				RGBD135 [72]				DUT-RGBD [73]				SIP [45]			
	$S_a \uparrow$	$F_\beta \uparrow$	$E_\xi \uparrow$	$\mathcal{M} \downarrow$	$S_a \uparrow$	$F_\beta \uparrow$	$E_\xi \uparrow$	$\mathcal{M} \downarrow$	$S_a \uparrow$	$F_\beta \uparrow$	$E_\xi \uparrow$	$\mathcal{M} \downarrow$	$S_a \uparrow$	$F_\beta \uparrow$	$E_\xi \uparrow$	$\mathcal{M} \downarrow$	$S_a \uparrow$	$F_\beta \uparrow$	$E_\xi \uparrow$	$\mathcal{M} \downarrow$	$S_a \uparrow$	$F_\beta \uparrow$	$E_\xi \uparrow$	$\mathcal{M} \downarrow$
DF [39]	.757	.742	.838	.141	.735	.744	.818	.151	.769	.682	.838	.099	.681	.573	.806	.132	.719	.748	.842	.150	.653	.673	.794	.185
CTMF [74]	.848	.771	.870	.086	.849	.779	.864	.085	.860	.723	.869	.056	.863	.778	.911	.055	.830	.790	.882	.097	.716	.684	.824	.139
MMCI [75]	.873	.829	.905	.068	.859	.803	.878	.079	.856	.729	.871	.856	.848	.762	.904	.065	.791	.751	.855	.113	.833	.795	.886	.086
TANet [76]	.871	.835	.916	.060	.878	.844	.909	.061	.886	.795	.916	.041	.858	.795	.919	.046	.808	.771	.866	.093	.835	.809	.894	.075
DMRA [73]	.752	.762	.816	.086	.886	.872	.921	.051	.899	.855	.942	.031	.900	.866	.944	.030	.888	.883	.930	.048	.800	.815	.858	.088
CPFP [77]	.879	.830	.903	.051	.878	.837	.895	.053	.888	.323	.924	.036	.872	.829	.927	.037	.749	.736	.815	.100	.850	.819	.899	.064
ICNet [78]	.903	.865	.915	.045	.894	.868	.905	.052	.923	.870	.944	.028	.920	.889	.959	.027	.852	.830	.897	.072	.854	.836	.899	.069
DCMF [79]	.883	.841	.904	.054	.889	.859	.897	.052	.900	.839	.933	.035	.877	.820	.923	.040	.798	.750	.848	.104	.859	.819	.898	.068
CoNet [44]	.905	.884	.927	.037	.895	.872	.912	.046	.908	.846	.934	.031	.911	.861	.945	.027	.919	.909	.948	.033	.858	.842	.909	.063
PGAR [80]	.903	.872	.917	.044	.906	.883	.914	.045	.918	.871	.948	.028	.894	.870	.935	.032	.920	.914	.944	.035	.875	.848	.908	.059
DANet [81]	.901	.868	.921	.043	.899	.871	.908	.045	.920	.875	.951	.027	.924	.899	.968	.023	.899	.888	.934	.043	.875	.855	.914	.054
JL-DCF [82]	.903	.869	.919	.040	.902	.885	.913	.041	.925	.878	.953	.022	.931	.900	.969	.020	.906	.882	.931	.043	.880	.873	.921	.049
SSF [83]	.887	.867	.921	.046	.899	.886	.913	.043	.914	.875	.949	.026	.905	.876	.948	.025	.916	.914	.946	.034	.868	.851	.911	.056
UC-Net [41]	.903	.885	.922	.039	.897	.889	.903	.043	.920	.890	.953	.025	.933	.917	.974	.018	.864	.856	.903	.056	.875	.868	.913	.051
A2dele [42]	.878	.874	.915	.044	.869	.874	.897	.051	.896	.878	.945	.028	.885	.865	.922	.028	.886	.890	.924	.043	.826	.825	.892	.070
S ² MA [84]	.890	.855	.907	.051	.894	.865	.896	.053	.915	.853	.938	.030	.941	.906	.974	.021	.903	.866	.921	.044	.872	.854	.911	.057
D ³ Net [45]	.899	.859	.920	.046	.901	.865	.914	.046	.912	.861	.944	.030	.898	.870	.951	.031	.775	.756	.847	.097	.860	.835	.902	.063
DRLF [85]	.888	.845	.915	.050	.886	.849	.901	.055	.903	.843	.936	.032	.895	.868	.954	.030	.826	.803	.870	.080	.850	.813	.891	.071
CDNet [86]	.896	.873	.922	.042	.885	.866	.911	.048	.902	.848	.935	.032	.875	.839	.921	.034	.880	.874	.918	.048	.823	.805	.880	.076
HAINet [87]	.907	.885	.925	.040	.912	.900	.922	.038	.924	.897	.957	.024	.935	.924	.974	.018	.910	.906	.938	.038	.880	.875	.919	.053
RD3D [88]	.911	.886	.927	.037	.916	.901	.918	.036	.930	.892	.958	.022	.935	.917	.975	.019	.913	.924	.949	.031	.885	.874	.920	.048
SPNet [89]	.907	.890	.926	.037	.925	.916	.923	.028	.927	.910	.958	.021	.945	.936	.976	.014	-	-	-	-	.894	.885	.919	.043
DSA ² F [90]	.897	.893	.927	.039	.904	.898	.922	.039	.918	.892	.950	.024	.916	.901	.955	.023	.921	.926	.950	.030	.862	.865	.908	.057
DCF [91]	.902	.891	.927	.039	.912	.902	.924	.035	.923	.891	.957	.021	.904	.893	.952	.024	-	-	-	-	.875	.875	.920	.051
TriTransNet [16]	.908	.893	.927	.033	.920	.919	.925	.030	.928	.909	.960	.020	.943	.936	.981	.014	.933	.938	.957	.025	.886	.892	.924	.043
BGRDNet [92]	.902	.895	.928	.036	.907	.908	.922	.034	.917	.902	.957	.023	.933	.931	.976	.016	.927	.935	.951	.029	.882	.889	.923	.045
DAIR [93]	.907	.894	.928	.036	.926	.917	.923	.029	.933	.908	.959	.020	.941	.934	.971	.016	-	-	-	-	.887	.889	.924	.047
GroupTransNet (ours)	.912	.898	.931	.030	.924	.923	.928	.027	.933	.912	.966	.016	.942	.939	.981	.012	.937	.941	.960	.022	.888	.897	.927	.040

where ϕ denotes the enhanced alignment matrix, W and H are the width and height of the image, respectively. MAE calculates the average pixel-wise absolute difference between the prediction and ground-truth, which is defined as:

$$\mathcal{M} = \frac{1}{W \times H} \sum_{i=1}^W \sum_{j=1}^H |P(i, j) - G(i, j)|, \quad (30)$$

where P denotes the prediction and G denotes the ground-truth. Moreover, we also plot the Precision-Recall (PR) curve and F-measure (F_β) curve.

4.1.3. Training protocols

For the sake of fairness, we follow the same mainstream protocols just like the previous works [45,80] to employ the first training setting relative to STEREO, NJU2K, NLPR, RGBD135 and SIP, and as in the previous works [42,44,73] to employ the second training setting relative to DUT-RGBD. When the validation is executed on the former, 1485 image pairs from NJU2K and 700 image pairs from NLPR are split for training, the remaining samples of these two and the entire samples of the other three are collected for testing. When the validation is executed on the latter, additional 800 image pairs from DUT-RGBD are split for training, and the remaining samples of it are collected for testing.

4.1.4. Implementation details

We implement all the experiments on four NVIDIA GeForce RTX 3090 GPUs with 24 GB memory using the PyTorch toolkit. The ResNet-50 [99] pretrained on the ImageNet [100] is used as the backbone of the proposed network. To avoid potential overfitting, image pairs in training set are augmented with random flipping, cropping and rotating. During both training and testing stages, the resolution of all RGB and depth input image pairs are resized to 256×256 . Our network is entirely end-to-end and does not use any pre-processing (e.g., HHA [101]) or post-processing (e.g., CRF [102]). The Adam [103] optimizer is adopted to optimize the network. The initial learning rate is set to $1e-4$, which is adjusted by multiplying by 0.1 after every 60 epochs. Our model is trained for 150 epochs with a batch size of 3. It takes about 15 h to train the model. The test time for each image pair is only 0.037 s.

4.2. Comparison with the state-of-the-arts

We compare the comprehensive performance of the proposed method with 27 state-of-the-art methods, which are DF [39], CTMF [74], MMCI [75], TANet [76], DMRA [73], CPFP [77], ICNet [78], DCMF [79], CoNet [44], PGAR [80], DANet [81], JL-DCF [82], SSF [83], UC-Net [41], A2dele [42], S²MA [84], D³Net [45], DRLF [85], CDNet [86], HAINet [87], RD3D [88], SPNet [89], DSA²F [90], DCF [91], TriTransNet [16], BGRDNet [92] and DAIR [93]. The saliency maps of these competitive methods are provided by the corresponding authors or generated by running their released source codes under the default parameters.

4.2.1. Quantitative comparison

The results of the quantitative comparison are listed in Table 1. As can be seen that our GroupTransNet is superior to all competitors in almost all metrics. Compared with the method TriTransNet, which also uses the transformer, the proposed method averagely achieves 1.8%, 2.3% and 1.9% performance gains on all datasets in terms of the metrics of S_a , F_β and E_ξ , respectively. In particular, the performance of this method compared with the method TriTransNet on NLPR datasets in terms of the metric of $\mathcal{M}$ is improved by 0.4%. In addition, the methods of using transformer generally have significantly better performance than the methods of not using transformer. These results thoroughly validate the high accuracy of our method, and its competitiveness compared to other comparative methods.

4.2.2. Qualitative comparison

The results of the qualitative comparison are shown in Fig. 6. As can be seen from it, GroupTransNet has the ability to accurately identify salient objects in a variety of scenarios, such as similar foreground and background, small objects, multiple objects, etc. In particular, TriTransNet, HAINet and CDNet may not highlight the object well, but the proposed method can highlight the clear structure of the whole completely and accurately, and maintain a high degree of similarity with the real results. The reason for this could be ascribed to the strengths of the proposed method in expressing heterogeneous modals and scales as well as contextual features in complex situations.

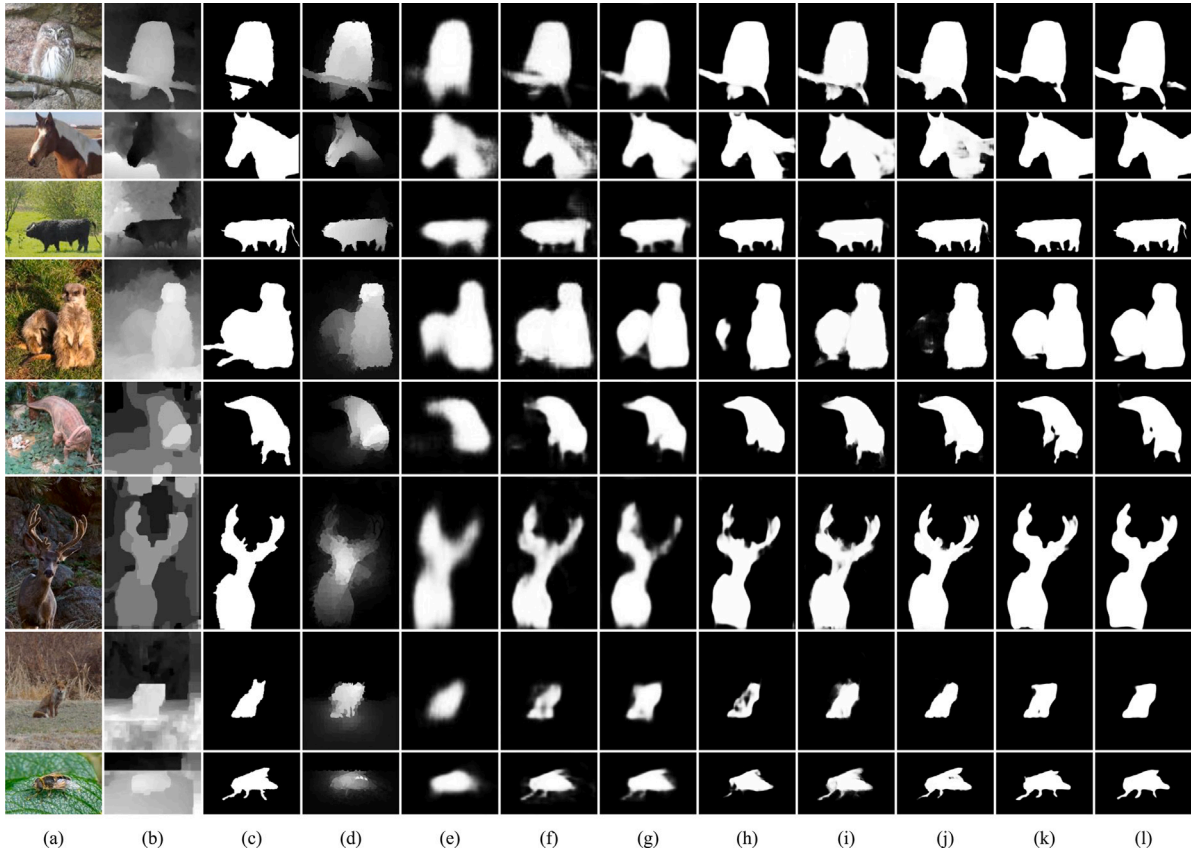


Fig. 6. Comparison of visualization results. (a) RGB; (b) Depth; (c) Ground-truth; (d) DF; (e) CTMF; (f) MMCI; (g) TANet; (h) DMRA; (i) D³Net; (j) HAINet; (k) TriTransNet; (l) The proposed method.

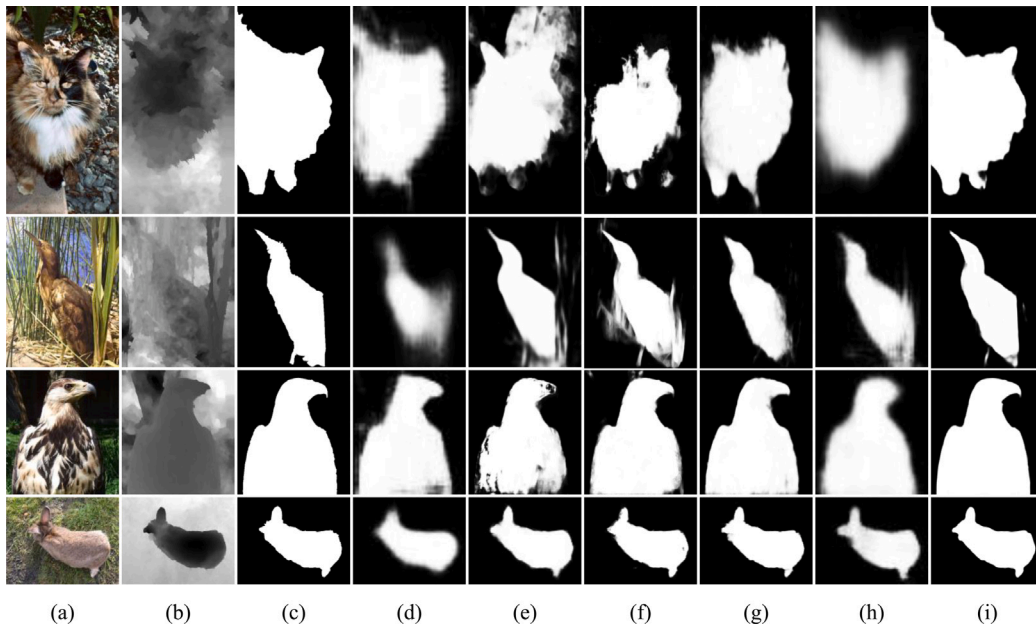


Fig. 7. Results of ablation experiments. (a) RGB; (b) Depth; (c) Ground-truth; (d) Baseline; (e) +MPM; (f) +SUM; (g) +MTE; (h) +CIU; (i) +All.

Table 2Ablation studies of MPM on NJU2K and SIP. The best result of each column is **bold**.

No.	Baseline	DEM	DPM	MPM	NJU2K				SIP			
					$S_\alpha \uparrow$	$F_\beta \uparrow$	$E_\xi \uparrow$	$\mathcal{M} \downarrow$	$S_\alpha \uparrow$	$F_\beta \uparrow$	$E_\xi \uparrow$	$\mathcal{M} \downarrow$
1	✓				.909	.904	.920	.035	.883	.887	.921	.044
2	✓	✓			.914	.910	.922	.033	.884	.889	.923	.044
3	✓		✓		.920	.919	.925	.030	.886	.892	.924	.043
4	✓			✓	.924	.923	.928	.027	.888	.897	.927	.040

Table 3Ablation studies of SUM on NJU2K and SIP. The best result of each column is **bold**.

Variants	NJU2K				SIP			
	$S_\alpha \uparrow$	$F_\beta \uparrow$	$E_\xi \uparrow$	$\mathcal{M} \downarrow$	$S_\alpha \uparrow$	$F_\beta \uparrow$	$E_\xi \uparrow$	$\mathcal{M} \downarrow$
Baseline	.909	.906	.919	.036	.881	.885	.920	.045
w/o SUM _H	.916	.915	.923	.032	.884	.890	.923	.043
w/o SUM _M	.919	.916	.924	.030	.885	.894	.924	.042
w/ SUM	.924	.923	.928	.027	.888	.897	.927	.040

Table 4Ablation studies of MTE on NJU2K and SIP. The best result of each column is **bold**.

Variants	NJU2K				SIP			
	$S_\alpha \uparrow$	$F_\beta \uparrow$	$E_\xi \uparrow$	$\mathcal{M} \downarrow$	$S_\alpha \uparrow$	$F_\beta \uparrow$	$E_\xi \uparrow$	$\mathcal{M} \downarrow$
Baseline	.889	.887	.914	.038	.868	.861	.896	.050
w/o MTE _H	.899	.898	.918	.034	.871	.869	.904	.046
w/o MTE _M	.904	.903	.921	.033	.873	.873	.908	.046
w/ MTE	.924	.923	.928	.027	.888	.897	.927	.040

4.3. Ablation analysis

To investigate the impact of each component of our proposed network on the performance of salient object detection, we conduct ablation experiments, and the visualization results are shown in Fig. 7. The baseline here refers to the regular network just like the feature pyramid networks (FPNs) [104]. It can be observed that as we separately assemble the proposed components onto the baseline, the predicted results are closer to the actual results, especially the contours become more concrete, indicating that the addition of each component has a certain positive effect. In order to further reveal the specific effect of each component, we carry out supplementary ablation experiments on NJU2K and SIP, involving four ablation studies. The baseline here generally refers to another set of processes, in which only the corresponding components in the proposed method are discarded.

4.3.1. Effect of MPM

Table 2 lists the results of ablation studies for MPM. The scheme of ‘No. 1’ used here is the baseline, which means that MPM is not applied. There are two other modules that are somewhat similar to MPM, namely, DEM [105] and DPM [16]. The schemes of ‘No. 2’, ‘No. 3’ and ‘No. 4’ represent adding them to the baseline, respectively. It can be seen that the embedding of MPM has an extraordinary significant effect on the improvement of performance. In addition, the performance improvement brought by the embedding of MPM is better than that of DEM or DPM.

4.3.2. Effect of SUM

Table 3 lists the results of ablation studies for SUM. Here, the scheme of ‘Baseline’ completely isolates the participation of SUM. Without the participation of SUM_H and SUM_M, they corresponding to the schemes of ‘w/o SUM_H’ and ‘w/o SUM_M’. The scheme of ‘w/ SUM’ represents the use of SUM. It can be seen that when removing SUM, it

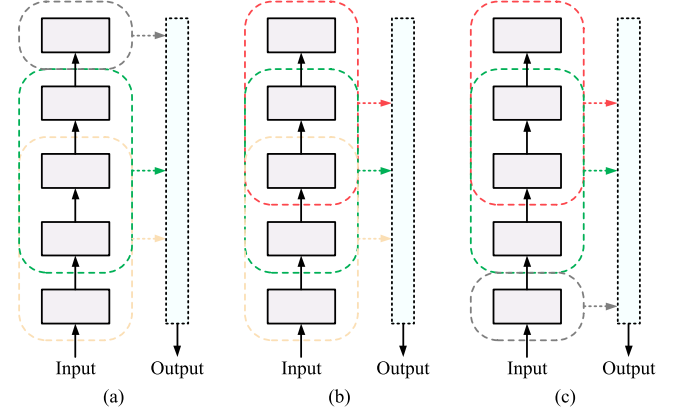


Fig. 8. Comparison diagram of different strategies on feature grouping. (a) Grouping for middle and low levels; (b) Grouping for all levels; (c) Grouping for high and middle levels.

can obtain $\mathcal{M}$ of 0.036 on NJU2K and $\mathcal{M}$ of 0.045 on SIP, respectively. This is continuously improved when SUM_H and SUM_M are deleted and SUM is used, respectively.

4.3.3. Effect of MTE

Table 4 lists the results of ablation studies for MTE. Here, the scheme of ‘Baseline’ completely isolates the participation of MTE. Without the participation of MTE_H and MTE_M, they corresponding to the schemes of ‘w/o MTE_H’ and ‘w/o MTE_M’. The scheme of ‘w/ MTE’ represents the use of MTE. It can be seen that compared with removing MTE, removing MTE_H and MTE_M both bring 0.4% improvement in terms of $\mathcal{M}$ on NJU2K, and bring 0.4% and 0.5% improvement in terms of $\mathcal{M}$ on SIP, respectively. When using MTE, the best results can be obtained.

4.3.4. Effect of CIU

Table 5 lists the results of ablation studies for CIU. The scheme of ‘No. 1’ used here is the baseline, which means that CIU is not applied. There are two other options different from CIU in feature fusion details. When all input features are fused together to produce an output, we name this method as overall merger unit (OMU). When these input features are fused according to the criteria of non-interleaving difference of features to generate three outputs, we name this method as corresponding coupling unit (CCU). The schemes of ‘No. 2’, ‘No. 3’ and ‘No. 4’ represent adding them to the baseline, respectively. It can be seen that with the embedding of OMU, CCU or CIU, the effect on performance improvement is getting better and better.

4.3.5. Different grouping strategies

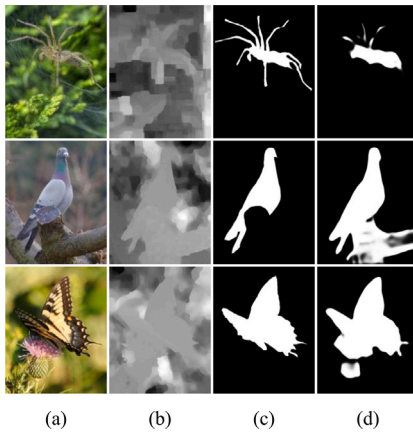
Fig. 8 illustrates different feature grouping strategies from input to output. In addition to the grouping of high and middle levels selected for use in the proposed method, there are also cases of grouping of middle and low levels, as well as all levels. For the grouping of middle and low levels, it is a mirrored form with the grouping of

Table 5Ablation studies of CIU on NJU2K and SIP. The best result of each column is **bold**.

No.	Baseline	OMU	CCU	CIU	NJU2K				SIP			
					$S_a \uparrow$	$F_\beta \uparrow$	$E_\xi \uparrow$	$\mathcal{M} \downarrow$	$S_a \uparrow$	$F_\beta \uparrow$	$E_\xi \uparrow$	$\mathcal{M} \downarrow$
1	✓				.915	.911	.917	.032	.881	.888	.917	.044
2	✓	✓			.917	.914	.922	.032	.882	.890	.919	.043
3	✓		✓		.919	.918	.924	.030	.885	.893	.924	.042
4	✓			✓	.924	.923	.928	.027	.888	.897	.927	.040

Table 6Ablation studies with different grouping strategies on NJU2K and SIP. The best result of each column is **bold**.

Strategies	Parameter (MB)	Speed (FPS)	NJU2K				SIP			
			$S_a \uparrow$	$F_\beta \uparrow$	$E_\xi \uparrow$	$\mathcal{M} \downarrow$	$S_a \uparrow$	$F_\beta \uparrow$	$E_\xi \uparrow$	$\mathcal{M} \downarrow$
Strategy <i>i</i>	738	47	.916	.919	.923	.029	.882	.889	.919	.043
Strategy <i>ii</i>	2098	19	.924	.925	.927	.027	.885	.893	.925	.041
Strategy <i>iii</i>	1285	34	.924	.923	.928	.027	.888	.897	.927	.040

**Fig. 9.** Some failure cases. (a) RGB; (b) Depth; (c) Ground-truth; (d) Ours.

high and middle levels. For the grouping of all levels, it reduces the isolated individual but increases one redundant grouping relative to grouping of high and middle levels. Table 6 lists the ablation results for different grouping. Here, ‘strategy *i*’, ‘strategy *ii*’ and ‘strategy *iii*’ refer to grouping for middle and low levels, all levels, and high and middle levels, respectively. It can be seen that ‘strategy *iii*’ represents the best compromise among all strategies in terms of parameter and speed of the model.

4.4. Failure cases

In Fig. 9, the failure cases of the proposed method in dealing with detection in some extreme environments are shown. We can observe that when the object is in a confusing state in the color map and depth map, or the information of the object in the depth map is not accurate enough, the detection performance of the method will deteriorate more or less. For example, in first line in the figure, the touch limbs of the spider are blurred in the color map and depth map, which cannot be well distinguished. The quality of the image of the second and third lines in the figure is very poor, so that its target is quite different from the shape of the pigeon and butterfly in the real picture. Therefore, there are some solutions that can be tried. For example, sorting the detected significant objects or intentionally deleting some invalid depth maps.

5. Conclusion

In this paper, we propose the group transformer network (GroupTransNet) for RGB-D salient object detection. GroupTransNet makes full use of transformer to learn the long-range dependencies of cross layer features at the lowest cost, thereby obtaining perfect feature expression between different levels of features. This method contains four basic components, namely, modal purification module (MPM), scale unification module (SUM), multiple transformer encoder (MTE) and cluster integration unit (CIU). Specifically, MPM realizes the matching of cross modal features by supplementing the mutual merits of color and depth modal, and feeding them via channel and spatial way. Through SUM, the higher and lower layers are sampled to the middle layer and processed to realize the unity and interrelation of the size of each scale feature. In MTE, different shared energy weights are placed to achieve the effects of high cohesion and low coupling of group. CIU is employed to cluster and integrate the features of different layers in a cascade manner. Compared with 27 state-of-the-art methods, the experimental results on six datasets in terms of four evaluation metrics confirm the superiority of the proposed method.

CRedit authorship contribution statement

Xian Fang: Writing – original draft, Visualization, Validation, Resources, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Mingfeng Jiang:** Writing – review & editing, Funding acquisition. **Jinchao Zhu:** Writing – review & editing, Visualization. **Xiuli Shao:** Writing – review & editing, Supervision. **Hongpeng Wang:** Writing – review & editing, Funding acquisition.

Declaration of competing interest

The authors declared that they have no conflicts of interest to this work. We declare that we do not have any commercial or associative interest that represents a conflict of interest in connection with the work submitted.

Data availability

Data will be made available on request.

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